Light before Dawn

Street lighting advances the first song of the day in four urban songbirds

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> *Before the light came, the blackbird had begun; it sang as though the lamp outside the window were the first edge of morning.*
>
> — FIELD NOTEBOOK, 14 APRIL 2024

ABSTRACT Songbirds time their first song of the day against the rising light, and the city adds light of its own. We analysed 4 545 species-mornings at 38 sites along a measured gradient of street lighting and estimated, for four common species, how far the onset of song moves for each tenfold increase in ground-level illuminance. Robins and blackbirds sing more than ten minutes earlier per decade of light, great tits about four, and chaffinches barely at all. The ordering follows eye size, which suggests that the response is set by how much light a species needs to see, not by its place in the chorus.

1 INTRODUCTION

The dawn chorus is among the most regular events in temperate ecology. Each species begins at a characteristic time before sunrise, and the order of entries is stable enough that field ornithologists once used it as a clock.[1] The cue behind that regularity is light: as the sky brightens through civil twilight, each species starts once illuminance crosses a threshold of its own.

Cities disturb the cue. A sodium or LED street lamp delivers several lux to a nearby perch through the whole night, which is brighter than the sky at the start of civil twilight. The best-known study of the effect found that males of several species close to street lights began singing earlier and fathered more extra-pair young (Kempenaers et al. 2010). Later work has shown that the spectrum of the lamp matters as much as its intensity[2] and that the size of a species' eye predicts how strongly it reacts.[3]

Most of this evidence comes from a handful of sites, compared as lit against unlit. Light at night is not a binary treatment, however; it falls off with distance and is shaded by trees and walls.[4] We therefore treat light as a continuous dose and ask a simpler question: how many minutes earlier does each species start for each tenfold increase in light?

1 P. Marler and H. Slabbekoorn (2004). *Nature's Music: The Science of Birdsong*. San Diego: Elsevier Academic Press.

2 R. Hale et al. (2017). "The Spectrum of Street Light and the Timing of Song in Urban Robins". In: *Journal of Avian Biology* 48.6, pp. 811–822.

3 C. Moreau et al. (2022). "Eye Size Predicts the Sensitivity of Dawn Song to Artificial Light across Forty European Passerines". In: *Proceedings of the Royal Society B* 289.1975, p. 20220391.

4 A lamp ten metres away delivers roughly a hundredth of the illuminance of one directly above the perch, so two territories on the same street can differ by two orders of magnitude.

Species	Sites	Mornings	β_j (min)	30 lux
Robin	34	1 288	−11.4 [−13.1, −9.7]	−17.0
Blackbird	36	1 344	−9.2 [−10.6, −7.8]	−13.7
Great tit	31	1 107	−4.1 [−5.3, −2.9]	−6.1
Chaffinch	22	806	−1.3 [−2.6, 0.0]	−1.9

TABLE 1 Shift in onset of song per tenfold increase in illuminance, from Equation (2). Negative values are earlier song. The last column is the predicted shift at 30 lux, a bright residential street.

2 METHODS

2.1 *Sites and light measurements*

We placed autonomous recorders at 38 sites across a city of about 300 000 people and its rural edge, chosen to span five orders of magnitude of night-time illuminance. At each site we measured horizontal illuminance at perch height with a calibrated lux meter on three moonless nights, following an established ground protocol,[5] and used the median as the site value L_i. Recording ran from 1 March to 31 May in two seasons.

5 I. da Costa and O. Whitfield (2019). "Light Pollution Gradients Measured from the Ground: A Calibrated Protocol for Ecologists". In: *Methods in Ecology and Evolution* 10.3, pp. 402–414.

2.2 *Defining the onset of song*

For each species, morning and site we counted songs per minute with a classifier trained on our own annotated recordings[6] and smoothed the counts with a five-minute running mean $r(t)$. The onset of song is the first time at which this rate reaches a fixed fraction of its maximum for that morning (Figure 1):

$$t^{*} = \min\bigl\{\, t : r(t) \geq \kappa\, r_{\max} \bigr\}, \qquad \kappa = 0.25. \tag{1}$$

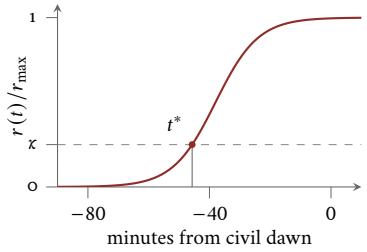


FIGURE 1 Onset t^* is the first minute at which the smoothed song rate reaches a fraction κ of its morning maximum.

A threshold relative to the morning maximum makes the onset insensitive to how far a bird sings from the recorder. We express t^* in minutes relative to the start of civil twilight, so that negative values mean song before civil dawn.

6 H. Verbeek et al. (2021). "Detecting the First Song of the Day in Continuous Field Recordings". In: *Ecological Informatics* 64, p. 101356.

2.3 *Model*

We fitted one linear mixed model per species j, with onset for morning i explained by the logarithm of site illuminance, day of the season D_i and a random intercept for site $s[i]$:

$$t^{*}_{ij} = \alpha_j + \beta_j \log_{10}\bigl(L_{s[i]} + 1\bigr) + \gamma_j D_i + u_{s[i]} + \varepsilon_{ij}, \qquad u_s \sim \mathcal{N}\bigl(0, \sigma_u^2\bigr)\,. \tag{2}$$

The slope β_j is the quantity of interest: the shift in onset, in minutes, for each tenfold increase in light. Adding one lux before the logarithm keeps unlit sites finite. Models were fitted in R,[7] and intervals are 95% profile likelihood intervals.

7 R Core Team (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna.

3 RESULTS

All four species began singing earlier at brighter sites, but by very different amounts (Table 1). Robins responded most strongly: at a site lit to 30 lux, a typical bright residential street, they started about seventeen minutes

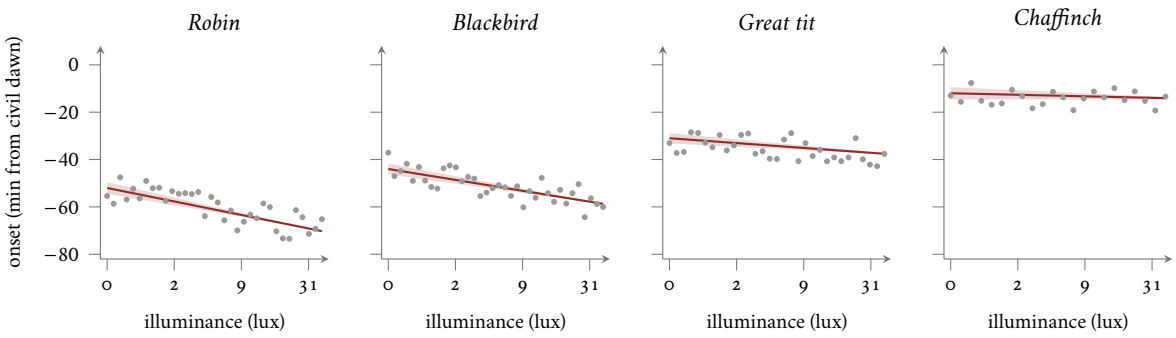


FIGURE 2 Site-mean onset of song against night-time illuminance, on the logarithmic scale of Equation (2). Lines are fitted slopes with 95% intervals; grey points are sites. The same axis is used for all species, so the panels read from left to right in the order of the chorus.

before robins at dark sites. Blackbirds were close behind. Great tits shifted by less than half as much, and the chaffinch slope was indistinguishable from zero.

The relation is close to linear on the logarithmic scale across the whole gradient (Figure 2). We found no sign of a threshold below which light has no effect: even the dimmest lit sites, at a fraction of a lux, sat above the dark-site mean for robins and blackbirds.[8]

SEASON. Onset advanced through the spring at a similar rate in all species, by 0.4 to 0.6 minutes a day, as civil dawn itself moved earlier. The light effect did not change with day of the season for any species, so the shift is not confined to the start of breeding.

SPECTRUM. Fourteen sites were converted from sodium to white LED lamps between the two seasons. Holding illuminance fixed, robins at converted sites started a further 3.8 minutes earlier [1.2, 6.4], consistent with the greater sensitivity of passerine vision to short wavelengths.[9]

8 A model with a breakpoint in $\log_{10} L$ did not improve on the linear one for any species (ΔAIC < 2).

9 R. Hale et al. (2017). "The Spectrum of Street Light and the Timing of Song in Urban Robins". In: *Journal of Avian Biology* 48.6, pp. 811–822.

4 DISCUSSION

The ranking of the four species matches the ranking of their eye size relative to body mass, which is largest in robins and smallest in chaffinches.[10] A large eye collects more light, so a species with a large eye sees well enough to forage and defend a territory earlier in twilight, and a street lamp moves that moment earlier still. The response would then be a matter of vision rather than of position in the chorus, which is the explanation favoured in a broad review of night-light impacts (Gaston et al. 2013).

Two cautions apply. Light and noise are correlated in cities, and although we measured traffic noise at every site, the two cannot be fully separated in an observational design. And an earlier start need not mean a longer day of song: whether lit birds also stop earlier in the evening is a question for the evening recordings, which we have not yet analysed.

10 C. Moreau et al. (2022). "Eye Size Predicts the Sensitivity of Dawn Song to Artificial Light across Forty European Passerines". In: *Proceedings of the Royal Society B* 289.1975, p. 20220391.

The practical message is simple. Because the effect is logarithmic, the first lux matters most: dimming a bright street from 30 to 10 lux recovers less of the dark-site timing than shielding a dark one from its first stray lamp.

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